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Name _____

April 12 , 2011 Sunday

Department _____

SID _____

University Physics
Sample Midterm Examination
School of Intensive Instruction in Sciences and Arts, Nanjing University

Physical Constants

Gravitational constant

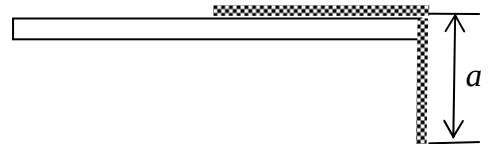
$$G = 6.67 \times 10^{-11} \text{ m}^3 / \text{s}^2 \cdot \text{kg}$$

Standard gravity acceleration

$$g_n = 9.806 \text{ m/s}^2$$

Select five out of following six problems.

(1) (20pts) As shown in the accompanying figure, a uniform chain of total length l hangs with a piece of length a ($0 < a < l$) over the edge of a plane table. The initial velocity of the chain is zero and there are no friction between the surface of the table and the chain.



- (a) Find the velocity of the chain at the instant it slides off the table.
- (b) Calculate the time in which the chain slides from the table.

(2) (20pts) The potential energy function for the force between two atoms in a diatomic molecule can be express as

$$U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$$

where a and b are positive constants and x is the distance between two atoms.

- (a) Find the force between two atoms as the function of x .
- (b) What is the equilibrium separation between two atoms?
- (c) Find the minimum energy needed to break the molecule apart.

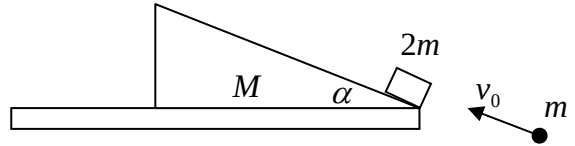
(3) (20pts) Assume that the Earth's orbit around the Sun is circular, and the Sun's mass is much larger than the Earth's mass.

(a) Compute the ratio of the kinetic and potential energies of the Earth.

(b) The Sun undergoes a supernova explosion and its mass suddenly reduces to half of its original value. Assume that the energy released in the explosion has no effect on the Earth. Compute the ratio of the kinetic and potential energies of the Earth at the instant after the explosion.

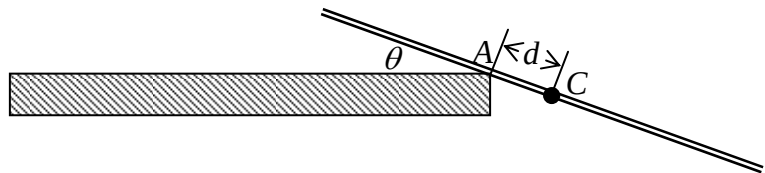
(c) What is the new total energy of the Earth? What is the shape of the orbit now? Will the earth escape the solar system?

(4) (20pts) A slider of mass $2m$ is initially at rest at the bottom edge of a wedge of mass M and angle α . The wedge has a frictionless surface and rests on a frictionless table. At $t = 0$, a bullet of mass m and velocity v_0 traveling parallel to the upper surface of the wedge hits and sticks in the slider.



- What is the maximum height reached by the slider?
- What is the wedge's velocity at the time the slider reaches its maximum height?
- How far has the wedge moved at that time?

(5) (20pts) A uniform rod of mass m and length l is held flat on a horizontal table and at right angle to an edge A of the table. The



center of mass C of the rod is beyond the edge of a distance d . The rod is released at rest and starts to rotate about the edge A and eventually slides off the table. The coefficient of static friction between the rod and the edge of the table is μ .

- Find the rotational inertias of the rod, I_C about the center of mass C and I_A about point A .
- Calculate the angular velocity of the rod as the function of rotation angle θ before sliding occurs.
- Determine the angle θ_0 when sliding of the rod begins.

(6) (20pts) A cylindrical wooden stick with lead attached at one end floats upright in water as shown in the figure. The total mass of the stick is m . In equilibrium, the length of submerged portion of the stick in water is l_0 . The density of the water is ρ and the area of the cross section of the stick is A .

- Write the equation of motion of the stick.
- Show that the oscillation of the stick is simple harmonic.
- Find the period of the oscillation.

